

Computational solution of wave problems

Project: FDTD simulation of diffraction over objects

Goal:

This project aims at getting insight in the finite difference time domain technique (FDTD). More specifically we will study the effect of perfectly matched layers and resolve the problem of discretisation of non-Cartesian surfaces in a Cartesian grid.

Exercise:

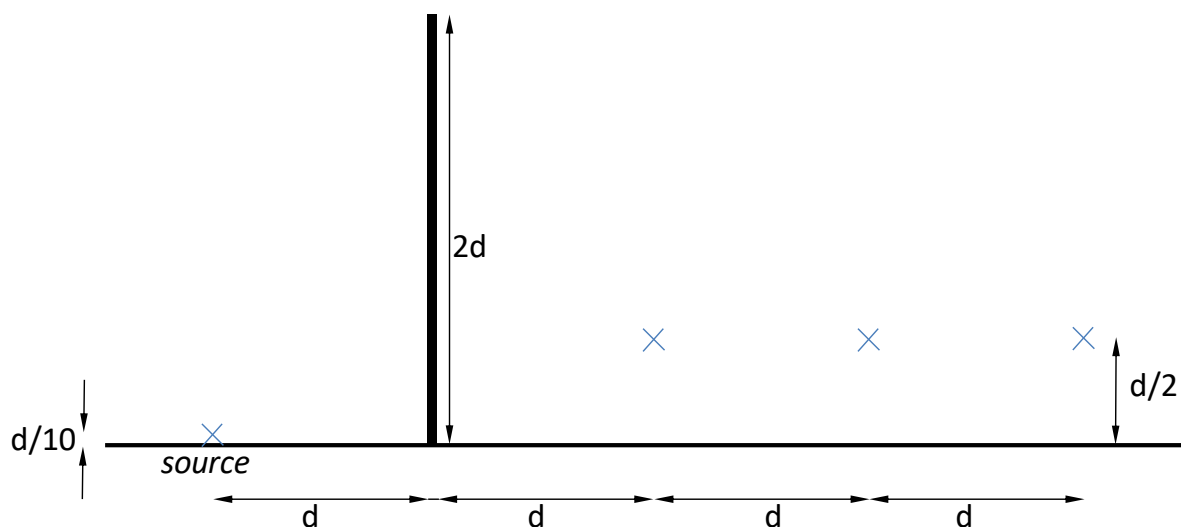
In this we consider a 2 dimensional problem, which implies that the objects are infinitely long in the direction orthogonal to the drawing and that the source is a line source. The source is transparent and infinitely small (p -source=monopole in 2 dimensions). The frequency range of interest is determined by $kd \in [0.1, 10]$.

Part 1: checking the implementation of PML using the example of an infinitely thin structure

The figure below shows the basic layout of the problem. The ground surface and all objects are perfectly hard: $\partial_n = 0$. The vertical screen has to be implemented to be as thin as possible (think carefully about this).

Implement a Cartesian grid FDTD with a non-reflecting perfectly matched boundary layer to approach the openness upward and to the left and the right. Calculate the effect of the ground plane and the screen on the propagation between the source and the 3 observation points marked with crosses in the figure below. To this end, obtain p at the three locations and divide it by p_0 obtained by FDTD at the same locations but without the ground and screen. One refers to this as the field relative to free field.

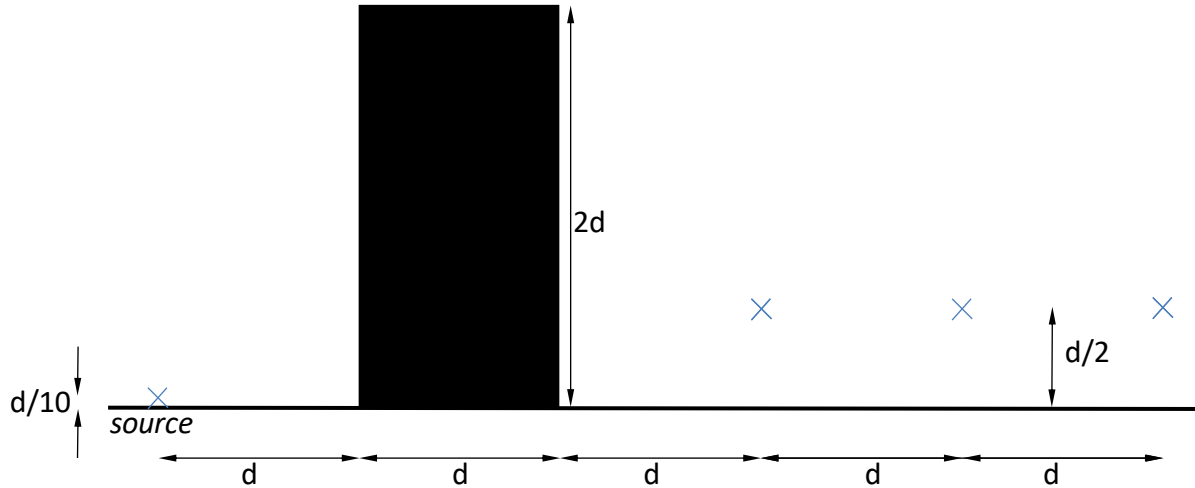
Compare this result with the analytical formulation for diffraction around a wedge that can be found in the chapter on geometrical methods (last chapter). (hint: use an image source to account for the reflecting ground surface in the analytical solution) Discuss the frequency range where the result is expected to be accurate from a theoretical point of view and based on the comparison with the analytical formulation. For this, start from the roughest grid that can describe the structure, the source, and the receiver and subsequently test at least one grid refinement. Consider the parameters used for implementing the PML carefully.



Part 2: extending to a thick object with finite surface impedance

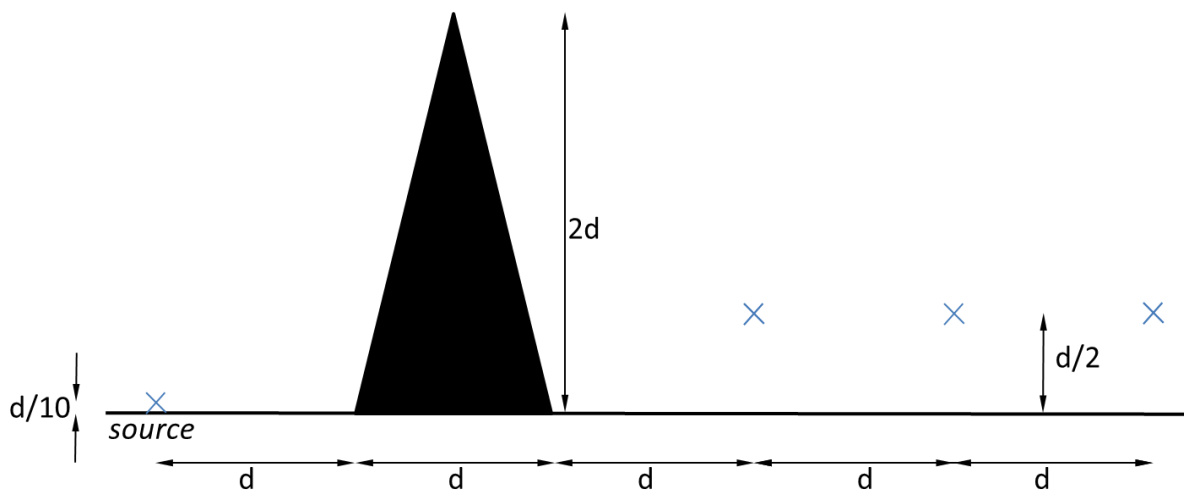
Now the central object is widened to become a thick object (figure below). Again calculate the field relative to free field in the 3 observation points and compare with the previous case. Note that the distance between source and receivers has changed compared to the situation in part 1.

Now implement a surface impedance $Z=2c$ on the object (not the ground). Explore its effect.



Part 3: resolving tilted boundaries

Now tilted boundaries are studied in the example of the triangular object shown below. Start by using the roughest Cartesian grid that you have chosen in Part 1. Then implement one of the techniques discussed in this course to more accurately model the non-Cartesian surface. Do this for both $\rho_n=0$ and $Z=2c$ on the surface of the wedge.



With this final model perform a thorough comparison of the frequency range (kd range) where you expect accurate results for different grid steps in a Cartesian grid and contrast to the CPU time needed to perform the calculation. This result obviously depends on the computer that you are using and the available memory.

Practical

Students work in groups of 3 for this project. The deliverable includes the software code and a discussion (report) of the results obtained with a focus on numerical aspects. The deadline for uploading the results via Ufora is one week before the exam. During the exam, the project will be discussed and additional theoretical questions will emerge from this.

Hints

- The developed code does not need to be universal. Choose optimal discretisation steps for describing this specific problem.
- Plot your results as a function of $k.d$ where k is the wave number $=2\pi/\lambda$ to make them independent of wave speed, c , and scalable with respect to distances and frequencies.